



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical
Sciences Chennai- 602105



Student Name : DIVYADHARSHINI P

Reg. No: 192312610

Course Code : CSA0204

Slot: A

Course Name :C-PROGRAMMING

Course Faculty: DR.A.SEVALKURAM

Final Assignment Title: SMART CAMPUS ENERGY MONITORING AND ENERGY-SAVING RECOMMENDATION SYSTEM USING C

Module Photograph:

```
C smart_campus.c X
C: > Users > sridh > Downloads > C smart_campus.c
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX_FACILITIES 10
5 #define DAYS 7
6
7
8 typedef struct {
9     char name[40];
10    float limit;
11    float energy[DAYS];
12    float total;
13    float average;
14    float savings;
15    float score;
16 } Facility;
17
18 void getInput(Facility f[], int n);
19
20 int main() {
21     Facility f[MAX_FACILITIES];
22     int n;
23     printf("\nSMART CAMPUS ENERGY MONITORING SYSTEM\n");
24     printf("Enter number of facilities (1-10): ", MAX_FACILITIES);
25     scanf("%d", &n);
26
27     // ... (rest of the code) ...
28
29     // ENERGY REPORT
30     printf("\n===== ENERGY REPORT =====\n");
31     printf("%-10s %-10s %-10s %-10s %-10s\n", "Facility", "Limit(kwh)", "Total(kwh)", "Average(kwh)", "Savings(kwh)", "Score(%)");
32     printf("Library      500.00      400.00       57.14       100.00       80.00\n");
33     printf("Computer Lab  800.00      825.00      117.86       -25.00        0.00\n");
34     printf("===== \n");
35     printf("Total Campus Energy Used : 1225.00 kwh\n");
36     printf("Overall Average per Facility per Day : 87.50 kwh\n");
37     printf("Program completed successfully.\n");
38     return 0;
39 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\sridh\Downloads> gcc smart_campus.c -o smart_campus
PS C:\Users\sridh\Downloads> .\smart_campus.exe

SMART CAMPUS ENERGY MONITORING SYSTEM
Enter number of facilities (1-10): 2

Enter facility 1 name: Library
Enter energy limit (kwh) for Library: 500
Enter energy used for day 1: 60
Enter energy used for day 2: 55
Enter energy used for day 3: 70
Enter energy used for day 4: 65
Enter energy used for day 5: 50
Enter energy used for day 6: 45
Enter energy used for day 7: 55

Enter facility 2 name: Computer Lab
Enter energy limit (kwh) for Computer Lab: 800
Enter energy used for day 1: 120
Enter energy used for day 2: 110
Enter energy used for day 3: 130
Enter energy used for day 4: 125
Enter energy used for day 5: 115
Enter energy used for day 6: 105
Enter energy used for day 7: 120

===== ENERGY REPORT =====
Facility      Limit(kwh)    Total(kwh)    Average(kwh)  Savings(kwh)  Score(%)
Library       500.00       400.00       57.14        100.00       80.00
Computer Lab  800.00       825.00      117.86       -25.00        0.00
=====
Total Campus Energy Used : 1225.00 kwh
Overall Average per Facility per Day : 87.50 kwh
Program completed successfully.
PS C:\Users\sridh\Downloads>
```

Ln 5, Col 15 Spaces: 4 UTF-8 CRLF {} C Win32

Project Description:

My primary contribution to the Smart Campus Energy Monitoring and Energy-Saving Recommendation System was focused on the design and implementation of the facility data management and input module. I contributed to designing the Facility structure used to represent the important information of each campus facility, including the facility name, facility type, permitted weekly energy limit, and seven days of energy consumption data. I worked on storing the daily energy readings using arrays within the structure and ensured that the collected information could be accessed and processed efficiently by the other modules of the program. I also contributed to the input and display functionality, allowing users to enter facility details and energy consumption values and view the stored records in a clear manner.

While developing these components, I applied C programming concepts such as structures, arrays, strings, functions, pointers, loops, conditional statements and formatted input/output. I also participated in organizing the program into separate functions so that the overall system remained modular, readable and easier to maintain. As part of my contribution, I tested the facility data entry and display operations using different sets of sample values, including normal consumption values and values near or beyond the permitted energy limit. I checked whether the entered data was stored correctly and whether the corresponding facility information and consumption values were displayed as expected.

I also assisted in testing the integrated program to ensure that the data-management components worked correctly with the other modules of the system, including analysis, searching and sorting. Through this project, I developed a better understanding of how structures and arrays can be used to model real-world data and how different C programming concepts can be combined to develop a practical application. I also gained experience in problem analysis, modular programming, testing, debugging and working collaboratively as part of a team. The project helped me understand how systematic monitoring of electricity consumption can support the identification of energy wastage and encourage responsible energy usage within a campus environment. Overall, my contribution helped establish the basic facility data-management and input components on which the remaining energy-monitoring and analysis functions of the system operate.

Student Signature**Guide Signature**